

Geophysical Research Letters[®]



RESEARCH LETTER

10.1029/2025GL119362

Key Points:

- The solar wind is unlikely to be the primary driver of the long-term cooling of Uranus's thermosphere
- Since 2009, the solar wind power at Uranus has increased, while in the same time-period Uranus's thermosphere has continued to decrease
- The decline in energy sourced from the solar wind is not sufficient to change the thermospheric temperature at Uranus

Correspondence to:

J. M. Jasinski,
jasinski@jpl.nasa.gov

Citation:

Jasinski, J. M., Melin, H., Sinclair, J. A., Zomerdijsk-Russell, S., Achilleos, N., & Paty, C. (2025). Uranus' long-term thermospheric cooling is unlikely to be primarily driven by the Solar wind. *Geophysical Research Letters*, 52, e2025GL119362. <https://doi.org/10.1029/2025GL119362>

Received 11 SEP 2025

Accepted 3 DEC 2025

Author Contributions:

Conceptualization: Jamie M. Jasinski

Data curation: Henrik Melin

Formal analysis: Jamie M. Jasinski, Henrik Melin, James A. Sinclair, Nick Achilleos, Carol Paty

Methodology: Jamie M. Jasinski, Henrik Melin, James A. Sinclair, Sophia Zomerdijsk-Russell, Nick Achilleos

Validation: Henrik Melin, James A. Sinclair

Visualization: Jamie M. Jasinski

Writing – original draft: Jamie M. Jasinski

Writing – review & editing: Jamie M. Jasinski, Henrik Melin, James A. Sinclair, Sophia Zomerdijsk-Russell, Nick Achilleos, Carol Paty

Uranus' Long-Term Thermospheric Cooling Is Unlikely to Be Primarily Driven by the Solar Wind

Jamie M. Jasinski¹ , Henrik Melin² , James A. Sinclair¹ , Sophia Zomerdijsk-Russell¹ , Nick Achilleos³ , and Carol Paty⁴ 

¹NASA Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA, ²Department of Mathematics, Physics, and Electrical Engineering, Northumbria University, Newcastle Upon Tyne, UK, ³Department of Physics and Astronomy, UCL, London, UK, ⁴Department of Earth Sciences, University of Oregon, Eugene, OR, USA

Abstract Over the last 40 yrs, Uranus' thermospheric temperature has decreased by ~8 K/yr. Here we show that this is less likely to be driven by solar wind kinetic power than previously thought. We determine that the solar wind kinetic power at Uranus has been increasing since the start of solar cycle 24 in 2009, while Uranus's thermospheric temperature has continued to decline. A comparison of the ascending phases of solar cycles 24 and 25 shows an increase in solar wind power of 15%, while Uranus's thermospheric temperature decreased by 18% in the same timeframe. If the solar wind was a driver of thermospheric temperature at Uranus it would be expected that Uranus would have reversed its declining temperature trend in recent years. We have also estimated that a decline in energy sourced from the solar wind would not be sufficient to change the thermospheric temperature by the level observed.

Plain Language Summary Since the 1980's until present day, the temperature of the upper atmosphere at Uranus has been steadily declining (i.e., cooling). The reason for this is currently unexplained. Multiple ideas have been put forth to explain this. This includes the solar wind as a driver of the temperature, since its power is also lower now compared to the 1980's. We show that since the start of solar cycle 24 in 2009 the solar wind power at Uranus has generally increased while the upper atmospheric temperature has continued to monotonically decline in the same timeframe, showing that they are not correlated. If the solar wind power did drive the temperature, then we would have expected a gradual increase in the temperature over the last ~16 yrs. We have also calculated that there is not enough energy lost from a decline in solar wind power to cool the upper atmosphere.

1. Introduction

In 1986, Voyager 2 completed a flyby of Uranus, and the Ultraviolet Spectrograph (UVS) onboard the spacecraft measured a H₂ exospheric temperature of 750 K at the top of the extended atmosphere (Broadfoot et al., 1986). This is hotter than the 400 K observed at Saturn and colder than the 1,450 K measured at Jupiter (Broadfoot et al., 1981; Sandel et al., 1982). H₃⁺ emission was detected from Uranus by ground based infrared telescopes, with an estimated temperature of 757 ± 25 K in the thermosphere (Trafton et al., 1993, 1999). Since then, multiple observations have shown that this measured H₃⁺ temperature has been steadily decreasing from 715 K in 1992 to 534 K in 2008, with an average decrease of 8 K/yr (Melin et al., 2011) which has continued at a similar cooling rate until present times (Melin et al., 2019). The same rate of cooling has been observed in H₂ observations (Trafton & Kaplan, 2025).

The explanation for this thermospheric cooling has remained elusive. Trafton et al. (1999) suggested this could be due to changes in the solar cycle. However, Melin et al. (2011) showed that there was no trend associated with the solar cycle. Uranus' exceptionally long orbital period of 84 yrs has also been suggested as a possible explanation for the temperature variation, however, this has not been demonstrated conclusively yet. McComas et al. (2013) reported that from the start of the space age until the 2010s there has been a gradual long-term decline in sunspot number, and that the observed solar wind steadily decreased in various measurable parameters, including dynamic pressure (decreased by ~41%) and energy flux (decreased by 48%). Masters et al. (2024) suggested that there could be a correlation of the declining temperature of the upper atmosphere with the decline of solar wind dynamic pressure (and therefore solar wind kinetic power) indicated by McComas et al. (2008, 2013).

© 2025. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Jasinski & Velli (2025) showed that the Sun has recently reversed this weakening trend, noting an increase of ~34% in dynamic pressure and an approximately ~40% increase in energy flux since 2008. However, Uranian exospheric temperatures have continued to monotonically decrease since 2008. In this paper, we examine the hypothesis presented by Masters et al. (2024) that the thermospheric temperature is driven by the solar wind. Firstly, we fit trends to the temperature and solar wind kinetic power to show that since the start of solar cycle 24 in 2009, the solar wind kinetic power at Uranus has increased while the temperature has continued to decrease. Secondly, we estimate possible energy budgets regarding the thermosphere and solar wind kinetic power.

2. Observations and Data Analysis of the Solar Wind

We use the OMNI-2 in-ecliptic long-term data set, that is, publicly available, which is an intercalibrated set of solar wind measurements made at ~0.99 au at the L1 Lagrange point by various spacecraft (King & Papitashvili, 2005; Papitashvili & King, 2020). Using this data set Jasinski & Velli (2025) showed that since 2008, the solar wind has reversed its weakening trend and that the solar wind proton dynamic pressure has increased from ~1.3 nPa in 2008 to ~1.8 nPa in 2025 at 0.99 au. This is still lower than the observed dynamic pressure of 2.26 nPa averaged over 1976–1989 (McComas et al., 2013).

Figure 1a shows the solar wind dynamic pressure. The data are averaged over a full solar rotation to produce the best whole-Sun averaged measurements and to reduce the scatter (Jasinski and Velli (2025); McComas et al., 2013). Figure 1b shows the mean value on a linear scale, with the expected values at Uranus's orbit of 19.2 au calculated using a simple propagation of ~1 au data to Uranus using a $1/r^2$ relation (that has been previously used by Jasinski et al., 2024; Masters et al., 2024).

Figure 1c shows the expected magnetopause (MP) subsolar location (R_{SS}) size of Uranus (black line) estimated from the following equation:

$$R_{SS} = \left(\frac{B_0^2}{2\mu_0 p_{dyn}} \right)^{1/\alpha} \quad (1)$$

where B_0 is the equatorial “surface” magnetic field strength of the planet at the 1 bar level (23,000 nT; Ness et al., 1986), and μ_0 is the permeability of free space. We use $\alpha = 5.7$, which was estimated from Voyager 2 (Jasinski et al., 2024) and has subsequently been used in recent magnetopause modeling (Zomerdijk-Russell et al., 2025; Zomerdijk-Russell & Jasinski, 2025).

Figure 1d shows the expected solar wind kinetic power at Uranus. The solar wind kinetic power is calculated using:

$$\text{Power} = p_{dyn} v_{SW} \pi R_{SS}^2 \quad (2)$$

where p_{dyn} is the solar wind momentum flux (or dynamic pressure), v_{SW} is the solar wind speed, and R_{SS} is the dayside magnetopause distance. This quantity multiplies the energy flux ($p_{dyn} v_{SW}$) incident on the projected cross-section of the area of the magnetosphere (πR_{SS}^2) which intercepts the solar wind flow. This equation represents the power obtained if all solar wind particles impacting the dayside magnetopause were to transfer their full energy. The mechanisms of this energy transfer will be described in Section 3.

Figure 1d also shows Uranus' thermospheric temperatures (red stars) measured from H_3^+ observations (Melin et al., 2011, 2019; Trafton & Kaplan, 2025). Figure 1d shows that solar wind power at Uranus modestly decreased from 1986 to the end of 2008, which also coincides with a decrease in temperatures in accordance with Masters et al. (2024). However, since the solar minimum in 2008–2009 the solar wind power has modestly increased while the temperatures have continued to decline.

Figure 2 focuses on the data set from the start of solar cycle 24 (at the end of 2008) onwards, when the solar wind power began to reverse its weakening trend. Figure 2a shows the total dynamic pressure in the same format as Figure 1a. Figures 2b and 2c show the solar wind power and the thermospheric temperatures. Figure 2 also shows a fitted trend to a linear model (i.e., $y = A + Bx$) by minimizing the chi-square error statistic on the average data

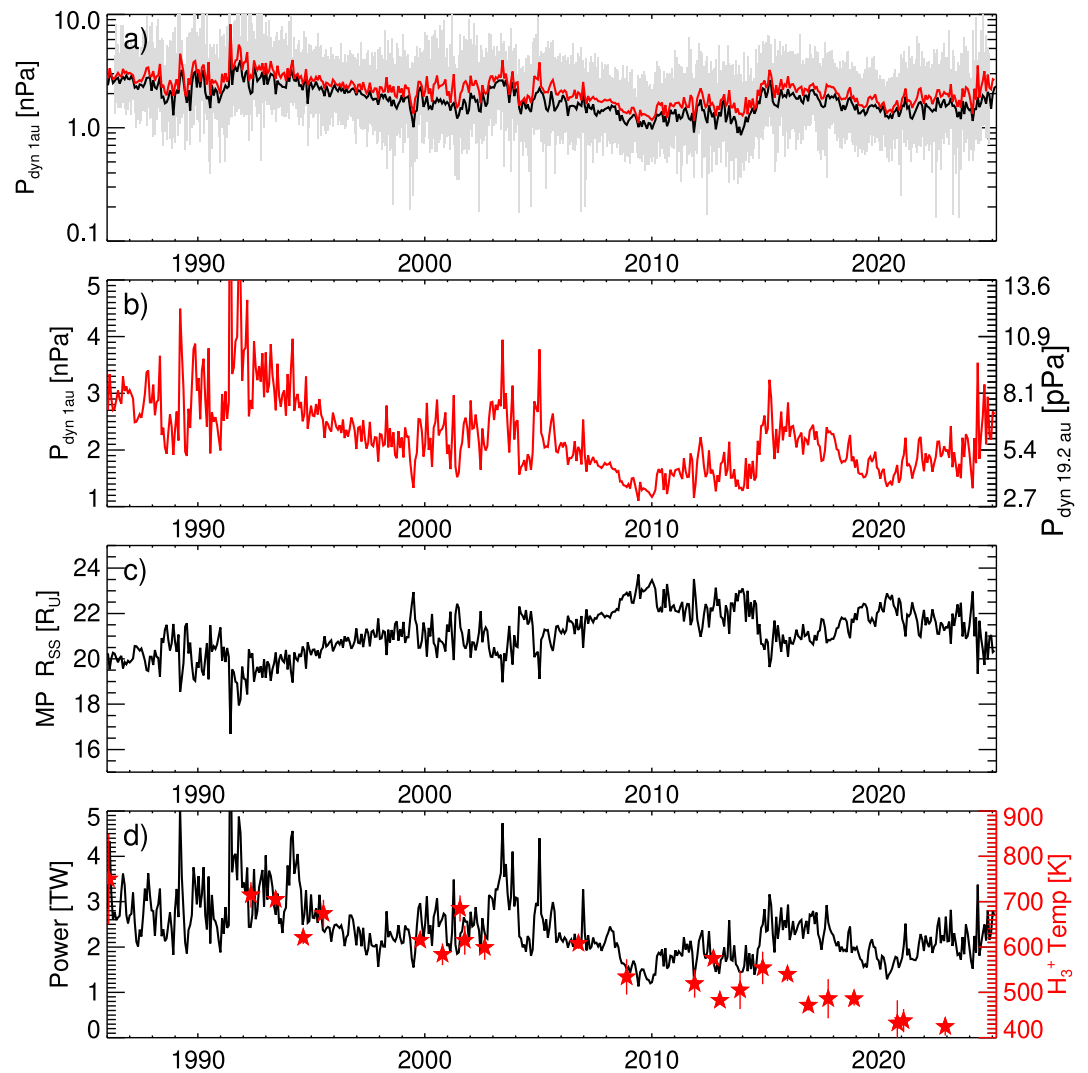


Figure 1. (a) Solar wind dynamic pressure at 1 au averaged over a solar rotation, red is the mean, black is median, and light gray show 5%–95% ranges; (b) mean dynamic pressure of the solar wind at 1 au (values on the left axis in nPa) and 19.2 au (values on right axis in pPa); (c) subsequent Uranian magnetopause size; and (d) solar wind power at Uranus and H_3^+ Uranian thermospheric temperatures.

(red line); using the same methodology as Jasinski & Velli (2025). The blue values in the bottom corners show the values of this fitted trend at the beginning and end of the timeseries.

Since the start of solar cycle 24, the trends show that the solar wind power at Uranus has increased by $\sim 29\%$ while the thermospheric temperature has decreased by $\sim 28\%$. We find a Pearson-correlation r -value of -0.1 with a p -value of 0.7 —that is, the Uranian thermospheric temperature is not correlated to the solar wind kinetic power since the Sun reversed its weakening trend (from the r -value) and that if there was a correlation it would not be stastically significant (i.e., the p -value). We have also calculated the Pearson-correlation coefficient for other time periods ending with the most recent temperature data. The temperature data end 3 yrs after the most recent solar minimum, so we calculated the correlation over two solar cycles starting at 3 yrs after a solar minimum (so that the data sets start and end at the same phase of a solar cycle; i.e. a like-for-like comparison). Aug 1999–Dec 2022 the correlation r -value for the two data sets is 0.3 with an p -value of 0.2 ; and for Nov 2012–Dec 2022 the correlation r -value is 0.2 with a p -value of 0.5 . This means there is no strong correlation between the two data set and that any weak correlation is statistically insignificant.

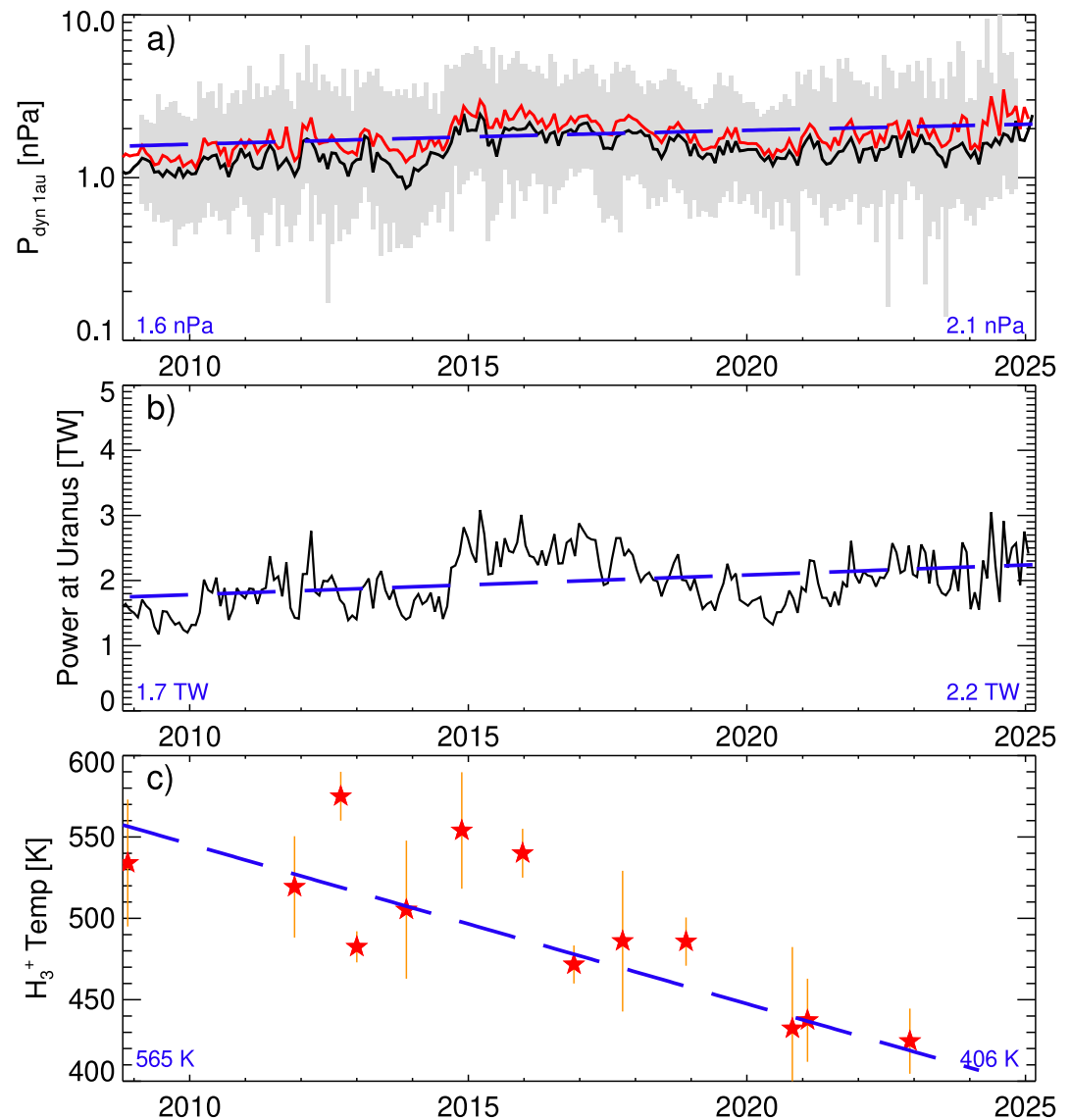


Figure 2. Starting from Oct 2008: (a) total dynamic pressure of the solar wind at 1au; (b) increasing solar wind power at Uranus and (c) decreasing Uranian thermospheric temperatures. Blue dashed line shows a fit to the data set, with the blue values representing the values at the start and end of the fitted trend.

We argue that recent changes in the Sun's activity show that the Sun has reversed its declining trend, while Uranus' thermospheric temperature has not. If the two data sets were indeed correlated, the temperature would not have continued to decline. It is expected that the thermospheric response time to solar wind variations is fast (~ 3 days; this is discussed in Section 4.1); so the temperature should have steadily increased over the last two solar cycles.

3. Energy Budget and Cooling Calculation

The coupling of a planet to its local space environment, and the associated transport of energy, occurs because the upper atmosphere and the solar wind are electromagnetically connected via the magnetosphere. This results in the transport of energy and momentum between the solar wind and the thermosphere due to global electrical current sheets which form field-aligned currents that focus and deposit energy to the thermosphere. These current systems are formed from various types of plasma flow within the magnetosphere but can be driven by different sources of energy. At Earth, for instance, the driver is the magnetospheric interaction with the solar wind (e.g., Birkeland & Aschehoug, 1908; Chapman & Ferraro, 1931, 1941; Cowley, 2000; Dungey, 1961). At Jupiter, however, plasma

Table 1
Comparison of Planetary and Magnetospheric Parameters (Adapted From Achilleos et al., 2021)

Planet/~rotation period t [h]	Relative magnetic moment/approx. Equatorial field [nT]	Solar wind travel time ^c T_S (Bow shock to planet), [min]	Ratio t/T_S	Average auroral brightness/ Input power	Global joule heating [GW]
Earth/24	1/30,000	2	720	1–100 kR (1–100 GW)	10–100
Jupiter/10	20,000/400,000	200	3	10–1,000 kR (10,000's GW)	~200,000
Saturn/11	580/20,000	40	16.5	1–100 kR (100–1000 GW)	~10,000
Uranus/17	50/23,000	24	43	1–2 kR ^a (30–70 GW) ^a	150–350 ^b

^aHerbert and Sandel (1994); Herbert (2009). ^bGershman and Clark (2025). ^cAssumes a Solar Wind Speed of 500 km/s.

flows are dominated by internal rotation imparted by the transport of angular momentum of the planet to the disc-like magnetosphere (e.g., Cowley & Bunce, 2001; Hill, 1979; Vasyliunas, 1983). Intense field-aligned currents are supported by motion of energetic particles along magnetic field lines, which can excite and ionize neutral particles in the atmosphere, provide local heating to the atmosphere, and affect atmospheric flows (e.g., Bagenal et al., 2017).

This system, which can transfer energy to the atmosphere from the magnetosphere, is not perfectly efficient. For example, typically, ~10% of energy carried by precipitating particles into the atmosphere forms auroral emissions (Achilleos et al., 2021). Table 1 summarizes some basic parameters for Earth, Jupiter, Saturn and Uranus. The table includes the planetary rotation period, t , the solar wind travel time from the bow shock to the planet, T_S , and a ratio of the two, t/T_S . This shows a basic comparison of whether magnetospheric dynamics is more likely to be driven by its interaction with the solar wind (i.e., a high ratio; e.g., Earth), or whether internal corotation is more likely to be the driver of dynamics (i.e., a low ratio; e.g., Jupiter). Also shown are the average auroral brightness and input power, as well as the estimated global joule heating of the planet's atmosphere from magnetosphere-ionosphere coupling. For all four planets, the global power dissipated by the atmospheric Joule heating (from ionospheric currents) is greater than the auroral energy input, showing that more energy from the magnetosphere goes into heating rather than generating the aurora.

We will now estimate how much energy needs to be lost from Uranus in order to produce the cooling of the thermosphere and compare this to the possible energy reduction from the decline in solar wind kinetic power. The hypothesis that the solar wind kinetic power drives the thermospheric temperature of Uranus (Masters et al., 2024) was motivated by the same coupling which powers radio emissions at the giant planets (i.e., Zarka et al., 2001).

For solar wind-magnetosphere coupling, there is a very well-defined relationship between the solar wind power and the auroral radio power across all five “radio planets.” Zarka et al. (2001) calculate the solar wind kinetic power using Equation 2 shown above. Equation 2, calculates the power obtained if all solar wind particles impacting the dayside magnetopause were to transfer their full energy. Uranus' radio power was found to be $\sim 10^7$ W, while the incident kinetic flow power is $\sim 10^{12}$ W, with a fitted efficiency of 7×10^{-6} (Zarka et al., 2001). This means that a very small fraction of the power of the solar wind is transported to the planet itself to produce the radio power. This is expected since the magnetosphere is a large volume of space with many plasma processes taking place that transfer energy from the solar wind, with a fraction of the solar wind kinetic power being imparted to the planet itself. This efficiency is estimated for radio power. However, a solar wind-magnetosphere coupling efficiency of ~1% is expected at Earth; that is, the overall coupling efficiency of solar wind kinetic power to Earth's magnetosphere is estimated to be ~1% (e.g., Stern, 1984; Østgaard et al., 2002). Others suggest as much as 4% (Lu et al., 1998).

Let's consider the solar wind power values that were reported by Masters et al. (2024), who wrote that “from ~1990 to ~2020 this power has dropped by ~50%, from ~4 to ~2 TW.” This corresponds to a change in energy of 0.066 TW per year (or 66 GW/yr). This is 2.1×10^3 J s⁻², which corresponds to an average change in incoming solar wind kinetic energy of $\sim 2 \times 10^{18}$ J every year. Multiplying this by the solar wind-magnetosphere coupling efficiency estimated for Earth, of 1%–4% (assuming all of this energy goes directly to the thermosphere) results in a decrease of ~ 2 – 8×10^{16} J of energy into the system per year (i.e., $\sim 10^{16}$ J/yr). This would be the reduction in energy from the declining solar wind power, which would result in the temperature cooling if the solar wind were the main driver of thermospheric temperature. An even higher and more liberal efficiency could be assumed by

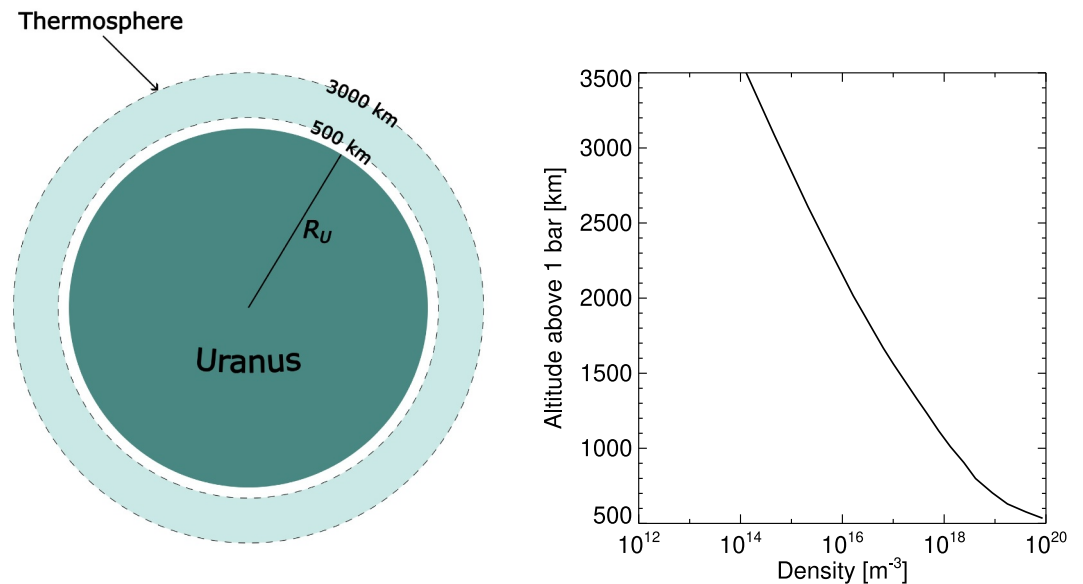


Figure 3. Modeled H_2 density above the 1 bar level at Uranus' thermosphere. Left: a schematic showing the thermospheric shell, that is, used to calculate the energy budget at Uranus's thermosphere. Right: the density altitude profile of H_2 at Uranus (taken from Moore et al., 2019).

comparing the total solar wind kinetic power of 2–4 TW to the total global joule heating 150–350 GW from Table 1, which would suggest an efficiency of 4%–18%. This would indicate a reduction of $8\text{--}36 \times 10^{16}$ J of energy per year from the solar wind at Uranus. Therefore, we would expect a change of solar wind kinetic power supplied to the thermosphere to be within a range of $10^{16}\text{--}10^{17}$ J per year, depending on the efficiency of solar wind-magnetosphere-thermosphere coupling. Given the paucity of data and knowledge of Uranus' magnetosphere we have used a wide range for the solar wind-magnetosphere-thermosphere coupling efficiency. We note that the liberal upper end of this range is extremely high, considering that at Earth the solar wind-magnetospheric coupling efficiency is expected to be 1%–4%, and this does not include coupling to the thermosphere itself from the solar wind (which would therefore produce an even lower solar wind-thermosphere coupling efficiency; i.e. this is the “solar wind-magnetosphere” coupling efficiency which would then have to be further lowered by multiplying by a “magnetosphere-thermosphere” efficiency).

How does the energy from the solar wind kinetic power compare to the amount of energy, that is, required to be taken out of the system to provide the reported change of ~ 8 K/yr (Melin et al., 2011, 2019) in thermospheric temperature at Uranus? Using the modeled density altitude profile shown in the right panel of Figure 3 (from Moore et al., 2019) of H_2 we estimate that there is approximately 2×10^{14} kg of molecular hydrogen in the thermosphere of Uranus. It is important to use the H_2 density profile since this is the dominant atmospheric component at Uranus at these altitudes, which would be in thermal equilibrium with the H_3^+ component (which by mass at Uranus is comparatively negligible to H_2). Observations of H_2 temperatures have also shown the same decline of temperature as H_3^+ (Trafton et al., 2005; Trafton & Kaplan, 2025). The energy (E) required to lower or raise the H_2 temperature is

$$E = C_{pH_2} n_{H_2} m_{H_2} \Delta T \quad (3)$$

where C_{pH_2} is the specific heat capacity of H_2 taken to be $14.5 \text{ kJ kg}^{-1} \text{ K}^{-1}$ at 500 K, n_{H_2} is the number of H_2 particles in the system, m_{H_2} is the mass of H_2 (i.e., a total of 2×10^{14} kg of H_2 at Uranus in a shell of volume from 500 to 3,500 km above the 1 bar level—see Figure 3) and ΔT is the temperature change of the gas (i.e., 8 K/yr).

The resulting energy change required to cool the H_2 thermosphere is 2×10^{19} J/yr, which we argue is a conservative estimate since we have not included other species which are present and will be in thermal equilibrium with H_2 . This energy is considerably higher than the 10^{16} J/yr energy that the solar wind power into the atmosphere decreases by every year assuming an Earth-like solar wind-magnetosphere coupling efficiency of 1%–4%.

Using the more liberal efficiency of solar wind-thermosphere coupling of 4%–18% (i.e. energy reduced from the declining solar wind of $8\text{--}36 \times 10^{16}$ J), is also not enough to cool the thermosphere. Therefore, we argue that the decades long decrease in solar wind power is not a viable explanation for the change in thermospheric temperature at Uranus (i.e., the reduction of energy is not enough to account for the energy required to cool the thermosphere).

We note that it is currently unknown whether H_2 and H_3^+ emissions at Uranus emanate from the same altitude layer. However, we do know that they emit at very similar altitudes at Jupiter (Uno et al., 2015). The long-term H_3^+ and H_2 temperatures are overall quite similar (Trafton & Kaplan, 2025), so this also supports similar altitudes for the temperature observations. Future, simultaneous H_2 and H_3^+ observations, would help to understand the short-term variations present within the longer-term cooling trend of Uranus's thermospheric temperature.

4. Discussion

4.1. Solar Wind Data Analysis

Jasinski & Velli (2025) showed that since 2008, the solar wind has reversed its weakening trend and that the solar wind dynamic pressure and energy flux have been modestly increasing. This led us to re-examine the idea that a continuing weakening of the solar wind at Uranus has caused the monotonic decrease in thermospheric temperature at the planet over the last 40 yrs (Masters et al., 2024). Over the last 16 yrs, however, the solar wind kinetic power at Uranus has increased from 1.7 TW to 2.2 TW, while the thermospheric temperature has continued its decline from 565 to 406 K in the same timeframe (Figures 2c and 2d). Recent JWST observations in 2023 have also shown the lowest temperatures on record (415 ± 13 K; Melin et al., 2025).

Figure 2 starts at solar minimum and ends during solar maximum, however we emphasize that the solar wind dynamic pressure and kinetic power increase occurs irrespective of the phase of the solar cycle. Jasinski & Velli (2025) showed the same increasing trend is still present when comparing the ascending phases of solar cycle 24 (2008.92–2013) and cycle 25 (2019.92–2024), that is, the same phase of a solar cycle. The authors found an increase in average dynamic pressure and energy flux of $\sim 18\%$ and $\sim 22\%$ respectively. Using the same methodology as Jasinski & Velli (2025) and McComas et al. (2013) when comparing the ascending phase of solar cycles 24 and 25, we find a change in the average solar wind kinetic power at Uranus of $+15\%$, and a change in the Uranian thermospheric temperature of -18% , between these two solar cycles ascending phase. This shows that when comparing the same phase of a solar cycle, there is still an increase in the average solar wind kinetic power while the thermospheric temperature continues to decrease.

The upper atmosphere of Uranus will also respond to changes in the overall energy balance (i.e., heating or cooling) with a delay which is dependent on the thermal inertia of the atmosphere. Whilst radiative timescales are very long in the lower atmosphere (Conrath et al., 1990), they are significantly shorter in the rarified thermosphere. Based on the heat capacity calculated above, the required energy to increase the temperature by 1 K is 0.3 kJm^{-2} . With an auroral Joule heating rate of $\sim 1 \text{ mWm}^{-2}$ (Gershman & Clark, 2025), and ignoring energy loss terms and dynamics, it would take ~ 3 days to change the thermal structure by 1 K. Therefore, we expect the time lag of the upper atmosphere in response to changing energy inputs to be very short when compared to the decades over which we have observed the long-term trends of the solar wind and the thermospheric temperature at Uranus.

We therefore suggest, that if the solar wind did drive Uranus' thermospheric temperature, then one would have expected a gradual increase in the temperature since the start of solar cycle 24 at the end 2008.

4.2. Energy Budget and Efficiency

We have estimated that there is approximately $\sim 10^{14}$ kg of H_2 at Uranus in a shell from 500 to 3,500 km above the 1 bar level, and that it would require $\sim 10^{19}$ J of energy to be lost to lower the temperature by approximately 8 K in a year. Using Equation 2 to estimate the of solar wind kinetic power at Uranus over the last 40 yrs, we estimate a reduction of $10^{16}\text{--}10^{17}$ J from the solar wind input at the thermosphere every year, with the large range dependent on the type of efficiency one assumes for solar wind-magnetosphere-thermosphere coupling.

Even if one assumes a 100% efficiency, the total energy reduction in the solar wind is $\sim 10^{18}$ J every year, which is also lower than the $\sim 10^{19}$ J/yr required to cool the thermosphere. Therefore, from an energy budget perspective, we suggest that longer-term solar wind variations are unlikely to be the driver of the longer-term cooling trend of

Uranus' thermospheric temperature. We do note that this is a simplistic energy estimation and that further modeling is required.

5. Conclusions

Uranus' thermospheric temperature has been monotonically decreasing over the last ~40 yrs, which has remained unexplained. McComas et al. (2013) showed that since the start of the space age the solar wind has been weakening, with a general decrease in the solar wind dynamic pressure. Masters et al. (2024) hypothesized that this decrease could be driving the decrease in Uranus' thermospheric temperature, and that the solar wind kinetic power could be a driver of the temperature.

Following up the recent study by Jasinski & Velli (2025), who demonstrated that since 2008 the solar wind has reversed the long-term weakening trend reported by McComas et al. (2013), we determine that the solar wind kinetic power at Uranus has increased by ~28% since the start of solar cycle 24 (at the end of 2008). If the solar wind is a driver of Uranus' thermospheric temperature, then one would have expected a gradual increase in the temperature since then. However, the temperature has continued to consistently decline over the same timeperiod. Considering the sparseness of the Uranian data set in comparison to short-term variations of the solar wind, this question remains to be resolved. Future observations, especially with the James Webb Space Telescope, will help resolve this question conclusively. However, we have also estimated that the change in energy required from the reduction in solar wind kinetic power at Uranus is most likely not enough to cause the cooling of its thermosphere.

Therefore, considering the current data sets and our energy budget calculation, we argue that the solar wind kinetic power is unlikely to be the primary driver of thermospheric temperature at Uranus.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

We acknowledge use of NASA/GSFC's Space Physics Data Facility's OMNIWeb service, and OMNI data. The OMNI data set provides solar wind parameters at 0.99 au which are publicly available at the OMNI database website at: <https://omniweb.gsfc.nasa.gov/>. Table 3 of Melin et al. (2019) lists annual global temperatures of Uranus' thermosphere, and Trafton & Kaplan (2025) published the latest temperatures which are publicly available at <https://arxiv.org/abs/2501.11298>. Table 1 is adapted from Achilleos et al. (2021).

Acknowledgments

JMJ and JAS acknowledges support from the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration, (contract 80NM0018 D0004). HM was supported by a STFC James Webb Fellowship (ST/W001527/2) at Northumbria University, UK. SZR is supported by an appointment to the NASA Postdoctoral Program at the Jet Propulsion Laboratory, California Institute of Technology, administered by Oak Ridge Associated Universities (ORAU) through a contract with the National Aeronautics and Space Administration (NASA). CP acknowledges support from the University of Oregon.

References

- Achilleos, N., Ray, L., & Yates, J. (2021). Magnetosphere–Ionosphere coupling. *Oxford Research Encyclopedia of Planetary Science*. Retrieved 25 Jul. 2025, from <https://oxfordre.com/planetaryscience/view/10.1093/acrefore/9780190647926.001.0001/acrefore-9780190647926-e-227>
- Bagenal, F., Adriani, A., Allegrini, F., Bolton, S. J., Bonfond, B., Bunce, E. J., et al. (2017). Magnetospheric science objectives of the Juno Mission. *Space Science Reviews*, 213(1–4), 219–287. <https://doi.org/10.1007/s11214-014-0036-8>
- Birkeland, E. P., & Aschehoug, H. (1908). On the cause of magnetic storms and the origin of terrestrial magnetism. In *The Norwegian Aurora Polar Expedition 1902–1903*, Vol. 1, 61–160.
- Broadfoot, A. L., et al. (1986). Ultraviolet spectrometer observations of Uranus. *Science*, 233, 74–79. <https://doi.org/10.1126/science.233.4759.74>
- Broadfoot, A. L., Sandel, B. R., Shemansky, D. E., McConnell, J. C., Smith, G. R., Holberg, J. B., et al. (1981). Overview of the Voyager ultraviolet spectrometry results through Jupiter encounter. *Journal of Geophysical Research*, 86(A10), 8259–8284. <https://doi.org/10.1029/JA086iA10p08259>
- Chapman, S., & Ferraro, V. C. A. (1931). A new theory of magnetic storms. *Terr. Magn.*, 36(2), 77–97. <https://doi.org/10.1029/te036i002p00077>
- Chapman, S., & Ferraro, V. C. A. (1941). The geomagnetic ring-current: Its radial stability. *Terr. Magn.*, 46, 1–6. <https://doi.org/10.1029/te046i001p00001>
- Conrath, B. J., Gierasch, P. J., & Leroy, S. S. (1990). Temperature and circulation in the stratosphere of the outer planets. *Icarus*, 83(2), 255–281. ISSN 0019-1035. [https://doi.org/10.1016/0019-1035\(90\)90068-K](https://doi.org/10.1016/0019-1035(90)90068-K)
- Cowley, S. W. H., & Bunce, E. J. (2001). Origin of the main auroral oval in Jupiter's coupled magnetosphere-ionosphere system. *Planetary and Space Science*, 49(10–11), 1067–1088. [https://doi.org/10.1016/S0032-0633\(00\)00167-7](https://doi.org/10.1016/S0032-0633(00)00167-7)
- Cowley, S. W. H. (2000). Magnetosphere–ionosphere interactions: A tutorial review, S. Ohtani, et al. In *Geophysical Monograph Series, Magnetospheric Current Sys-tems*, (Eds.), (Vol. 118, pp. 91–106). AGU <https://doi.org/10.1029/GM118p0091>
- Dungey, J. W. (1961). Interplanetary magnetic fields and the auroral zones. *Physical Review Letters*, 6(2), 47–48. <https://doi.org/10.1103/physrevlett.6.47>
- Gershman, D. J., & Clark, G. (2025). Predictions of auroral electron precipitation and Joule heating at Uranus. *Geophysical Research Letters*, 52(9), e2024GL114470. <https://doi.org/10.1029/2024GL114470>
- Herbert, F. (2009). Aurora and magnetic field of Uranus. *Journal of Geophysical Research*, 114(A11), A11206. <https://doi.org/10.1029/2009JA014394>

- Herbert, F., & Sandel, B. R. (1994). The Uranian aurora and its relationship to the magnetosphere. *Journal of Geophysical Research*, 99(A3), 4143–4160. <https://doi.org/10.1029/93JA02673>
- Hill, T. W. (1979). Inertial limit on corotation. *Journal of Geophysical Research*, 84(A11), 6554–6558. <https://doi.org/10.1029/JA084iA11p06554>
- Jasinski, J. M., Cochrane, C. J., Jia, X., Dunn, W. R., Roussos, E., Nordheim, T. A., et al. (2024). The anomalous state of Uranus's magnetosphere during the Voyager 2 flyby. *Nature Astronomy*, 9(1), 66–74. <https://doi.org/10.1038/s41550-024-02389-3>
- Jasinski, J. M., & Velli, M. (2025). The Sun reversed its decades-long weakening trend in 2008. *The Astrophysical Journal Letters*, 990(2), L55. <https://doi.org/10.3847/2041-8213/adb3a6>
- King, J. H., & Papitashvili, N. E. (2005). Solar wind spatial scales in and comparisons of hourly wind and ACE plasma and magnetic field data. *Journal of Geophysical Research*, 110(A2). <https://doi.org/10.1029/2004JA010649>
- Lu, G., Baker, D. N., McPherron, R. L., Farrugia, C. J., Lummerzheim, D., Ruohoniemi, J. M., et al. (1998). Global energy deposition during the January 1997 magnetic cloud event. *Journal of Geophysical Research*, 103(11), 11685–11694. <https://doi.org/10.1029/98ja00897>
- Masters, A., Szalay, J. R., Zomerdijk-Russell, S., & Kao, M. M. (2024). Solar wind power likely governs Uranus' thermosphere temperature. *Geophysical Research Letters*, 51(22), e2024GL111623. <https://doi.org/10.1029/2024GL111623>
- McComas, D., Angold, N., Elliott, H., Livadiotis, G., Schwadron, N., Skoug, R., & Smith, C. (2013). Weakest solar wind of the space age and the current "mini" solar maximum. *The Astrophysical Journal*, 779(1), 2. <https://doi.org/10.1088/0004-637x/779/1/2>
- McComas, D. J., Ebert, R. W., Elliott, H. A., Goldstein, B. E., Gosling, J. T., Schwadron, N. A., & Skoug, R. M. (2008). Weaker solar wind from the polar coronal holes and the whole Sun. *Geophysical Research Letters*, 35(18), 2008GL034896. <https://doi.org/10.1029/2008gl034896>
- Melin, H., Fletcher, L. N., Hammel, H. B., Milam, S. N., Moore, L., Stallard, T. S., et al. (2025). The ionosphere of Uranus as revealed by JWST. *Geophysical Research Letters*, 52(22), e2025GL118301. <https://doi.org/10.1029/2025GL118301>
- Melin, H., Fletcher, L. N., Stallard, T. S., Miller, S., Trafton, L. M., Moore, L., et al. (2019). The H 3 + ionosphere of Uranus: Decades-long cooling and local-time morphology. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 377, 2154. (20180408). <https://doi.org/10.1098/rsta.2018.0408>
- Melin, H., Stallard, T., Miller, S., Trafton, L. M., Encenaz, T., & Geballe, T. R. (2011). Seasonal variability in the ionosphere of Uranus. *The Astrophysical Journal*, 729(2), 134. <https://doi.org/10.1088/0004-637x/729/2/134>
- Moore, L., Melin, H., O'Donoghue, J., Stallard, T. S., Moses, J. I., Galand, M., et al. (2019). Modelling H+ 3 in planetary atmospheres: Effects of vertical gradients on observed quantities. *Phil. Trans. R. Soc. A*, 377(2154), 20190067. <https://doi.org/10.1098/rsta.2019.0067>
- Ness, N. F., Acuña, M. H., Behannon, K. W., Burlaga, L. F., Connerney, J. E. P., Lepping, R. P., & Neubauer, F. M. (1986). Magnetic fields at Uranus. *Science*, 233(4759), 85–89. <https://doi.org/10.1126/science.233.4759.85>
- Østgaard, N., Germany, G., Stadsnes, J., & Vondrak, R. R. (2002). Energy analysis of substorms based on remote sensing techniques, solar wind measurements, and geomagnetic indices. *Journal of Geophysical Research*, 107(A9), 1233. <https://doi.org/10.1029/2001JA002002>
- Papitashvili, N. E., & King, J. H. (2020). OMNI hourly data [DataSet]. *NASA Space Physics Data Facility*. <https://doi.org/10.48322/1shr-ht18>
- Sandel, B. R., Shemansky, D. E., Broadfoot, A. L., Holberg, J. B., Smith, G. R., McConnell, J. C., et al. (1982). Extreme ultraviolet observations from the voyager 2 encounter with Saturn. *Science*, 215(4532), 548–553. <https://doi.org/10.1126/science.215.4532.548>
- Stern, D. P. (1984). Energetics of the magnetosphere. *Space Science Reviews*, 39(1–2), 193–213. <https://doi.org/10.1007/BF00173674>
- Trafton, L. M., Geballe, T. R., Miller, S., Tennyson, J., & Ballester, G. E. (1993). Detection of H 3 + from Uranus. *The Astrophysical Journal*, 405, 761. <https://doi.org/10.1086/172404>
- Trafton, L. M., Melin, H., Geballe, T. R., & Stallard, T. S. (2005). On the mapping efficiency of integral field vs Longslit Spectrographs: Comparison of the UIST/IFU and CGS4 at UKIRT using Uranus in the K & L bands technical report, on the United Kingdom Infrared Telescope web site (pp. 1–11).
- Trafton, L. M., Miller, S., Geballe, T. R., Tennyson, J., & Ballester, G. E. (1999). H2 quadrupole and H+3 emission from Uranus: The uranian thermosphere, ionosphere, and Aurora. *The Astrophysical Journal*, 524(2), 1059–1083. <https://doi.org/10.1086/307838>
- Trafton, L. M., & Kaplan, K. ~F. (2025). High spectral resolution observations of Uranus' near-IR thermospheric H2 emission spectrum using the IGRINS spectrograph during the 2018 and 2023 apparitions. *Icarus*, 429, 116453. <https://doi.org/10.1016/j.icarus.2024.116453>
- Uno, T., Kasaba, Y., Tao, C., Sakanoi, T., Kagitani, M., Fujisawa, S., et al. (2015). Vertical emissivity profiles of Jupiter's northern H3+ and H2 infrared auroras observed by Subaru/IRCS. *Journal of Geophysical Research: Space Physics*, 119(10), 219. <https://doi.org/10.1002/2014JA020454>
- Vasyliūnas, V. M., & Press. (1983). Plasma distribution and flow. In *Physics of the Jovian magnetosphere* (pp. 395–453). Cambridge University. <https://doi.org/10.1017/CBO9780511564574.013>
- Zarka, P., Treumann, R. ~A., Ryabov, B. ~P., & Ryabov, V. ~B. (2001). Magnetically-Driven planetary radio emissions and application to extrasolar planets. *Astrophysics and Space Science*, 277(1–2), 293–300. <https://doi.org/10.1023/A:1012221527425>
- Zomerdijk-Russell, S., & Jasinski, J. M. (2025). Magnetopause reconnection voltages during the Voyager 2 flyby of Uranus. *Geophysical Research Letters*, 52(22), e2025GL117958. <https://doi.org/10.1029/2025GL117958>
- Zomerdijk-Russell, S., Jasinski, J. M., & Masters, A. (2025). Variation of model-predicted reconnection voltages applied to Uranus' dayside magnetosphere. *Journal of Geophysical Research: Space Physics*, 130(6), e2025JA033834. <https://doi.org/10.1029/2025JA033834>